Supply Chain Management Analysis Report

Interactive Power BI Dashboard Performance & Documentation Summary

Scope: Departmental Performance & Logistics Audit **Date:** July 2026

Active Dashboard Filter Context: Region: **West** | Category: **Furniture**

Dashboard Overview

The Supply Chain Management Analysis Dashboard provides an interactive, data-driven visualization of critical supply chain performance indicators. Designed for executive decision-makers and operations managers, it facilitates comprehensive monitoring across warehouse utilization, inventory efficiency, regional transportation costs, sales performance, and order fulfillment workflows.

By integrating real-time KPIs with cross-filtering slice-and-dice functionality, the report helps evaluate multi-regional operational efficiency and pinpoint logistical bottlenecks requiring strategic remediation.

Key Performance Indicators (KPIs)

WAREHOUSE UTILIZATION

32.39%

Measures current storage occupancy against total warehouse volumetric capacity limits.

DAYS SALES OF INVENTORY

16.38 Days

Average number of days inventory remains in stock before transitioning to a finalized sale.

INVENTORY TURNOVER

22.17

Frequency rate at which absolute stock levels are sold and replenished over the active period.

KPI Operational Interpretation: A DSI of 16.38 days coupled with an Inventory Turnover Ratio of 22.17 reflects high liquidity and robust inventory velocity. However, the Warehouse Utilization rate of 32.39% indicates significant unutilized capital capacity in storage infrastructure, presenting a strategic opportunity for consolidation or scaling.

Dashboard Visualizations & Detailed Observations

1. Warehouse Utilization Gauge

The gauge chart visualizes absolute warehouse capacity usage on a 0% to 100% linear threshold scale. Currently tracking at **32.39%**, the metric formally flags underutilization. This visual signal informs management that existing infrastructure can handle an influx of product lines without incurring overhead expansion costs.

2. Transportation Cost by Region and Category

This column visualization breaks down aggregated shipping and logistics expenses across geographic jurisdictions. Under the current filter configuration, the **West region** exhibits the highest concentrated transportation cost for the **Furniture** category, marking it as a critical focus area for regional freight optimization and carrier negotiations.

3. Units Sold Trends (Year-over-Year)

A longitudinal line chart tracks total unit volume trends over multi-year cycles, detailing market demand dynamics:

Reporting Year	Units Sold (Volume)	Performance Trend Context
2020	~ 2.0K units	Baseline establishing initial operational footprint.
2022	~ 14.6K units	Historical peak volume, reflecting major demand acceleration.
2023	~ 13.8K units	Marginal contraction (-5.4%), indicating early market stabilization.
2024	~ 11.5K units	Continued volume deceleration, signaling shifting buyer dynamics.

4. Average Lead Time by Category

Represented via a donut chart layout, this metric assesses supplier responsiveness and logistics fulfillment velocity. The average lead time for **Furniture stands at 15.33 days**. While reasonable for bulky goods, reducing this metric through streamlined regional dispatch could drastically increase localized customer satisfaction scores.

5. Order Fulfillment & Backorders Status

A distribution bar chart tracks pipeline order accuracy and structural operational strain based on total lifecycle status:

Order Lifecycle Status	Order Volume Count	Fulfillment Contribution Share
Fulfilled	60 orders	72.3% — Confirms strong downstream operations.
Pending	21 orders	25.3% — Represents active pipeline or short-term constraints.
Canceled	2 orders	2.4% — Exceptionally low rate, indicating order precision.

6. Inventory Level by Category & Region

This horizontal bar chart provides stock status visibility across localized supply networks. For the **Furniture** class within the **West region**, an active volume of approximately **0.2 million units** is maintained, ensuring adequate fulfillment coverage without causing over-saturation or inventory stagnation.

Synthesized Business Insights

- **Infrastructure Slack:** Low warehouse utilization (32.39%) suggests that capital is tied up in unutilized physical square footage. Management should explore subleasing storage space or diversifying product lines to fill capacity.
- **High Capital Velocity:** The combined inventory metrics (16.38 DSI and 22.17 Turnover) prove that products move exceptionally fast through the distribution network once received, mitigating cash-flow blockage risks.
- **Logistical Disproportionality:** Transportation costs are highly concentrated in the West territory for Furniture lines. A targeted freight audit is advised to discover whether this is driven by fuel surcharges, suboptimal carrier mix, or last-mile challenges.
- **Demand Posture Adjustments:** The volume decline from 14.6K in 2022 to 11.5K in 2024 requires a recalibration of safety stock limits to ensure warehouse space is not over-allocated to slowing product variants.

Conclusion

The Supply Chain Management Analysis Dashboard serves as a powerful decision-support framework by synthesizing disparate data streams into actionable KPIs. By highlighting a high-turnover but under-utilized storage model, the data supports an operational pivot toward regional transportation cost controls and strategic volume expansion. Incorporating these multi-layered filters ensures data integrity and agility across changing supply chain environments.